An Inverse-Arctangent Function Is Not Stieltjes

Moment rigidity, bounded phases, and an exact Hankel certificate

Keio Research Notes

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Abstract

We give a negative answer to the Stieltjes possibility raised in Question 1 of Keady, Khajohnsaksumeth, and Wiwatanapataphee (2018). The positive inverse of $z \mapsto \arctan(1/z)/z$ is not a Stieltjes function. The main proof uses positivity alone: three asymptotic coefficients would force a representing measure to have zero variance, hence to be a point mass, whereas an exact strict inequality rules out the resulting rational function. We extend the obstruction to bounded monotone phases with a nonnegative quadratic coefficient. Independently, we exhibit a strictly negative 4×4 normalized derivative Hankel determinant, with its sign certified by rational polynomial arithmetic. The complete-monotonicity questions about this inverse and its square are not settled here.

1 The published question and the conclusion

G. Keady, N. Khajohnsaksumeth, and B. Wiwatanapataphee, *On functions and inverses, both positive, decreasing and convex: And Stieltjes functions*, Cogent Mathematics & Statistics 5:1 (2018), article 1477543, DOI 10.1080/25742558.2018.1477543, define

$$\varphi_1(z) = \frac{\arctan(1/z)}{z}, \qquad \varphi_2(z) = \varphi_1(\sqrt{z}), \qquad z > 0.$$

Question 1, on printed page 2 of 16, asks:

> Is the inverse of φ_2 completely monotone? Is the inverse of φ_1 completely monotone (and, perhaps even Stieltjes)?

On printed page 13, Section 5.2, the authors specifically state that they have no obstruction to $\mu = \varphi_1^{-1}$ being Stieltjes. They already know that $\mu_{(2)} = \varphi_2^{-1} = \mu^2$ is not Stieltjes. The result below settles only the parenthetical Stieltjes possibility for μ. It does not prove or disprove complete monotonicity of either μ or μ^2.

Theorem 1. *The positive inverse $\mu = \varphi_1^{-1}$ is not a Stieltjes function.*

2 Rigidity from three asymptotic coefficients

If a Stieltjes function F satisfies

$$F(x) = \frac{c}{x+d} + o(x^{-3}) \qquad (x \to \infty), \quad c > 0, \quad d \geq 0,$$

then $F(x) = c/(x+d)$ for every $x > 0$. To prove this, write its Stieltjes representation, absorbing any pole at zero into the positive measure. Since $F(x) \to 0$, its additive constant is zero, and

thus

$$F(x) = \int_{[0,\infty)} \frac{d\rho(t)}{x+t}.$$

The first of the following three limits, followed by monotone convergence, gives $\rho([0,\infty)) = c$. Having established this, the second limit gives the finite first moment cd. With that moment established, the third limit gives the finite second moment cd^2:

$$\begin{aligned} xF(x) &= \int \frac{x}{x+t}\, d\rho(t) \longrightarrow c, \\ x^2\left(\frac{c}{x} - F(x)\right) &= \int \frac{xt}{x+t}\, d\rho(t) \longrightarrow cd, \\ x^3\left(F(x) - \frac{c}{x} + \frac{cd}{x^2}\right) &= \int \frac{xt^2}{x+t}\, d\rho(t) \longrightarrow cd^2. \end{aligned}$$

Each integrand is nonnegative and increases to the corresponding moment integrand. No finiteness of a higher moment has been assumed. It follows that

$$\int (t-d)^2\, d\rho(t) = cd^2 - 2d(cd) + d^2 c = 0.$$

Positivity forces $\rho = c\delta_d$, proving the lemma.

3 The inverse-arctangent obstruction

The derivative

$$\varphi_1'(z) = -\frac{\arctan(1/z)}{z^2} - \frac{1}{z(1+z^2)} < 0$$

and its endpoint limits show that φ_1 is a smooth decreasing bijection of $(0,\infty)$ onto itself. Put $L = \pi/2$. The inverse relation is

$$y\mu(y) = L - \arctan \mu(y), \qquad y > 0.$$

It gives $0 < \mu(y) < L/y$. Since

$$0 < u - \arctan u = \int_0^u \frac{t^2}{1+t^2}\, dt < \frac{u^3}{3} \qquad (u > 0),$$

we obtain the exact, strict, all-y estimate

$$0 < \mu(y) - \frac{L}{y+1} = \frac{\mu(y) - \arctan \mu(y)}{y+1} < \frac{L^3}{3y^3(y+1)}. \tag{R}$$

In particular $\mu(y) = L/(y+1) + O(y^{-4})$. If μ were Stieltjes, the moment-rigidity lemma would force equality with $L/(y+1)$ everywhere, contradicting the strict lower inequality in (R). This proves the theorem without numerical computation or any derivative beyond the one needed to define the inverse.

3.1 Positive scale parameters

For every $\alpha, b > 0$, the positive inverse of

$$\varphi_{\alpha,b}(z) = \alpha \frac{\arctan(b/z)}{z}$$

is not Stieltjes. Indeed it is $g_{\alpha,b}(y) = b\mu(by/\alpha)$, and the Stieltjes class is invariant under positive multiplication and positive scaling of the argument in either direction. In the Robin normalization $\varphi_\beta(z) = \arctan(1/(\beta z))/z$ with $\beta > 0$, this inverse is $\beta^{-1}\mu(y/\beta)$, so the negative conclusion applies to every positive Robin parameter. The limiting elementary inverse at $\beta = 0$ is L/y, which is Stieltjes.

4 A necessary condition for bounded phases

Let $L > 0$ and let $q : [0, \infty) \to [0, L)$ be nonzero, with $q(0) = 0$, $q \in C^1(0, \infty)$, a C^2 extension at zero, and $q'(z) \geq 0$ for $z > 0$. Define

$$\Phi(z) = \frac{L - q(z)}{z}, \qquad z > 0.$$

Then Φ is a strictly decreasing bijection of $(0, \infty)$ onto itself. If its inverse $u = \Phi^{-1}$ is a Stieltjes function, it is necessary that

$$\boxed{q''(0) < 0.}$$

In particular, $q''(0) \geq 0$ rules out the Stieltjes property.

Proof. The strict derivative inequality

$$\Phi'(z) = \frac{-zq'(z) - L + q(z)}{z^2} < 0$$

and the endpoint limits establish the bijection. Write $q(z) = az + bz^2 + o(z^2)$, where $a = q'(0) \geq 0$ and $b = q''(0)/2$. The inverse identity $yu(y) = L - q(u(y))$, initially giving $0 < u(y) \leq L/y$, yields successively

$$u(y) = \frac{L}{y} - \frac{aL}{y^2} + \frac{a^2L - bL^2}{y^3} + o(y^{-3}).$$

If u is Stieltjes, exactly the three monotone-convergence steps in the rigidity lemma give a finite positive representing measure with moments

$$m_0 = L, \qquad m_1 = aL, \qquad m_2 = a^2L - bL^2.$$

Its squared deviation from a is

$$\int (t - a)^2 \, d\rho(t) = -bL^2 \geq 0,$$

so $b \leq 0$. If $b = 0$, the measure must be $L\delta_a$, hence $u(y) = L/(y+a)$. Substitution into the inverse relation, using that u ranges over all of $(0, \infty)$, forces $q(z) = az$ for every $z > 0$. Boundedness of q forces $a = 0$, and then $q \equiv 0$, contrary to the hypotheses. Thus $b < 0$.

The result applies, under the displayed monotonicity and smoothness hypotheses, to every bounded nonzero odd phase: its second derivative at zero vanishes. In particular it applies to $q(z) = \arctan z$ with $L = \pi/2$, and to $q(z) = \tanh z$ or $q(z) = \operatorname{erf} z$ with $L = 1$. The result for these additional phases is a derived extension; no publication-priority claim is made.

The strict necessary sign is nonvacuous. For example, $q(z) = Lz/(1+z)$ has $q''(0) = -2L < 0$, and its inverse is Stieltjes:

$$u(y) = \frac{\sqrt{1 + 4L/y} - 1}{2} = \frac{1}{2\pi} \int_0^{4L} \frac{1}{y + t} \sqrt{\frac{4L - t}{t}} \, dt.$$

The equality follows by $t = 4L \sin^2 \theta$. This example does not assert that the sign condition is sufficient for arbitrary phases.

5 An independent finite Hankel certificate

The following exact local certificate independently proves the same theorem and records how a finite moment-Gram test detects the obstruction. Set

$$a = \frac{\pi}{4}, \qquad \tau = \frac{1}{1 + 2a} = \frac{2}{2 + \pi}, \qquad s_n = \frac{(-1)^n}{n!} \mu^{(n)}(a).$$

In particular $\mu(a) = 1$. Differentiating $y\mu(y) = \arctan(1/\mu(y))$ gives the exact differential identity

$$[1 + y(1 + \mu^2)]\mu' + \mu + \mu^3 = 0. \tag{A}$$

Substitution of the Taylor series at a into (A) yields

$$\begin{aligned}
s_0 &= 1,\\
s_1 &= 2\tau,\\
s_2 &= 4\tau^2 + 2\tau^3,\\
s_3 &= 8\tau^3 + \frac{32}{3}\tau^4 + 4\tau^5,\\
s_4 &= 16\tau^4 + \frac{112}{3}\tau^5 + \frac{100}{3}\tau^6 + 10\tau^7,\\
s_5 &= 32\tau^5 + \frac{1624}{15}\tau^6 + \frac{488}{3}\tau^7 + 112\tau^8 + 28\tau^9,\\
s_6 &= 64\tau^6 + \frac{1416}{5}\tau^7 + \frac{9128}{15}\tau^8 + \frac{6272}{9}\tau^9 + 392\tau^{10} + 84\tau^{11}.
\end{aligned}$$

These expressions can be checked independently, without (A), by inverting the Taylor series of $(\pi/2 - \arctan(1 + h))/(1 + h)$. Its first six nonconstant coefficients, multiplied by 2τ, are

$$-1, \quad 1 + \frac{\tau}{2}, \quad -1 - \frac{2\tau}{3}, \quad 1 + \frac{2\tau}{3}, \quad -1 - \frac{37\tau}{60}, \quad 1 + \frac{23\tau}{40}.$$

The companion exact-arithmetic script verifies this composition and the following determinant identity as polynomial identities over $\mathbb{Q}[\tau]$:

$$\det(s_{i+j})_{i,j=0}^{3} = \frac{8}{2025}\tau^{15}P(\tau), \qquad P(t) = -96+1280t-4540t^2-2820t^3+8100t^4+8100t^5+2025t^6. \tag{B}$$

Only $3 < \pi < 4$ is needed for its sign: this gives $1/3 < \tau < 2/5$. Put $u = 15\tau - 5 \in (0, 1)$. Direct expansion gives

$$P\left(\frac{u+5}{15}\right) = -\frac{1279}{9} - \frac{562}{9}u + \frac{283}{45}u^2 + \frac{1232}{225}u^3 + \frac{37}{75}u^4 + \frac{2}{125}u^5 + \frac{1}{5625}u^6 \le -\frac{730334}{5625} < 0.$$

Alternatively, in the Bernstein basis $\binom{6}{j}u^j(1-u)^{6-j}$, the seven coefficients have common denominator 16875 and numerator vector

$$(-2398125, -2573750, -2742300, -2899155, -3039140, -3156480, -3244752),$$

another exact certificate of strict negativity.

For any Stieltjes function f and any $a > 0$, its normalized signed derivatives $m_n = (-1)^n f^{(n)}(a)/n!$ are moments of a finite positive measure. Indeed, from

$$f(y) = c_0 + \frac{c_1}{y} + \int_{[0,\infty)} \frac{d\sigma(t)}{y + t}, \qquad c_0, c_1 \ge 0, \quad \sigma \ge 0,$$

the constant is an atom of mass c_0 at 0, the pole is an atom of mass c_1/a at $1/a$, and the remaining measure is the pushforward of $(a + t)^{-1}d\sigma(t)$ under $t \mapsto (a + t)^{-1}$. All moments exist since the support is contained in $[0, 1/a]$. Thus every Hankel matrix (m_{i+j}) is positive semidefinite: for a real vector v, its quadratic form is $\int(\sum_i v_i r^i)^2 d\nu(r) \ge 0$. The strictly negative determinant in (B) contradicts this necessary condition for μ to be Stieltjes, providing a second proof.

6 Source audit and exact limits of the claim

The publisher's indexed PDF directly supplies Question 1 on page 2 and Section 5.2 on page 13. The HTML publisher record gives publication on 14 June 2018 (the PDF also records the earlier accepted-version date 29 May 2018). Searches for the specific inverse-arctangent Stieltjes question, including Keady plus Hankel/determinant terms and searches restricted to the Pudim repository, found no earlier proof of the theorem above. This is not an exhaustive publication-priority certification.

The public Pudim application ledger entry APP-0035 concerns the different Question 3, about arbitrary completely monotone self-bijections. It gives a counterexample of the form $x^{-1} + 100e^{-x}$; that result does not address $\mu = \varphi_1^{-1}$. The full rendered application ledger was searched: it contains no occurrence of `arctan`, and its Keady entry concerns only Question 3. It does not claim the specific Stieltjes result proved here. Its full large TeX theory file could not be retrieved through the available public-fetch tool, so no claim of an exhaustive full-repository audit is made.

The 2025 Jovanović–Treml paper on the *reciprocal* $1/\arctan x$ concerns a different function; it is not a result about the compositional inverse φ_1^{-1}.

References and source links

- Publisher paper: Keady–Khajohnsaksumeth–Wiwatanapataphee (2018), publisher record
- Publisher PDF: Article 1477543, publisher PDF
- Related Robin-eigenvalue paper, definitions and context: `https://files.ele-math.com/articles/mia-21-62.pdf`
- Public competing-claim ledger: Pudim application ledger (APP-0035, GitHub)